

Welcome to Survey of Calculus! Please read through this syllabus carefully. If you have any questions or concerns, please ask!

Please see your Blackboard Course for information about your instructor.

REQUIRED MATERIALS

• MyLab Math

Homework, quizzes, unit exams, and the final exam will be completed online using MyLab Math(MLM). MLM can be accessed through your university Blackboard account. An electronic copy of the textbook, *Calculus with Applications*, Lial/Greenwell/Ritchey, 12th edition, is available in the MLM software

• PointSolutions Clicker

The point solutions audience response system will be used during class for routine in-class points. For more information, visit <http://clickers.uark.edu>. You may use the physical clicker device OR the PointSolutions mobile app. You will need to register your clicker device.

• Calculator

You will need a calculator. Any ACT approved calculator may be used for homework and in-class assignments. However, TI-84 is highly recommended. You will want to be familiar with the TI-84 as this is the calculator that will be provided during an exam.

For all tests that are taken in the Testing Lab, you will only be permitted to use one of the Math Department calculators. You should be familiar with this calculator as the proctors cannot answer questions about the operation of the calculator during a quiz, test, or the final exam.

• Internet Connection

Access to a reliable internet connection is required for this course. A problem with your internet may not be used as an excuse for late, missing, or incomplete coursework. If you experience problems with your internet connection while working on this course, it is your responsibility to find an alternative internet access point, such as a public library or Wi-Fi hotspot.

PROBLEMS WITH MLM SOFTWARE

If you can open your course but you are not able to access your MLM assignments, have tried a few times and still receive some type of an error message, then call the Technical Support line at 888-883-1299. Be sure that your browser will support the software. **SURE TO RUN THE BROWSER CHECK ONCE YOU GET STARTED!!).** F

Course Objectives

Upon successful completion of this course, you will be able to:

- CO1 Determine the existence of and evaluate limits using graphs, tables, and algebraic methods.
- CO2 Apply the appropriate rules to find derivatives of algebraic, exponential, and logarithmic functions.
- CO3 Associate the instantaneous rate of change and the slope of the tangent line with the derivative.
- CO4 Evaluate integrals using basic rules and the substitution method.
- CO5 Analyze functions and their graphs using limits, derivatives, and integrals.
- CO6 Apply the appropriate tools of calculus to solve applied problems in business, social science, and life sciences.

receive messages that your session has timed out. This often is due to the cookies downloaded to your computer from the MML website. To resolve this issue, either delete the cookies from your computer or try logging in with a different browser (e.g. Chrome, Firefox, etc.).

SYLLABUS QUIZ

Your first assignment in MyLab Math (MLM) is a syllabus quiz. You must score 100% on the syllabus quiz before you can begin working on any other assignments in MyLab Math and so you are encouraged to read through the syllabus carefully. If you have questions about the syllabus or the syllabus quiz content, please contact your instructor.

HOMEWORK

Unit homework assignments will be completed in MLM. You may work on each assignment to improve your grade until it is due by clicking on the “MyLab Math Homework” link in Blackboard. After the assignment is due, MLM will allow you to review your homework at any time without changing your grade by clicking on the “MyLab Math Gradebook” link in Blackboard.

QUIZZES

All MLM quizzes are due on **TUESDAY** nights and may be taken anywhere you have a reliable internet connection. You must earn the required percentage on each homework assignment in order to take the corresponding quiz. You may take each quiz up to three times and your highest score will count toward your overall final grade. Be aware that the questions may change for each attempt. Therefore, it is best to study before taking the quiz. There are no make-ups for missed quizzes and *your instructor cannot give partial credit on any quiz problems.*

Note: Each unit practice exam will count as a quiz grade in the MyLab Math gradebook. You will have unlimited attempts for each practice exam and your highest attempt will count toward your grade.

You are responsible for making sure your internet/computer is functioning properly. Do not wait until the last minute to take your quizzes. To accommodate illness, technical difficulties, etc., at least one quiz grade will be dropped at the end of the semester.

CLICKER GRADE

Our purpose for using Clickers during class time is to give both you and your instructor instant feedback on how well you understand and apply the material being covered in real time. Your instructor will attempt to provide clarity after Clicker questions, but it is also the responsibility of the student to use these opportunities to ask questions about answers you got wrong. This is much more than an attendance grade but **you must be physically present in class to earn credit.** Clicker questions will be administered at various times during each class period and may be related to current class material or review material. Points earned (by answering correctly!) during each class period will count toward your clicker grade. Clicker points cannot be earned while attending a class section for which you are not enrolled. Your clicker quiz grade accounts for 8% of your overall course grade.

You may use the physical clicker device or the PointSolutions mobile app in class. If you are using the physical clicker device, you must register your clicker device to receive clicker credit. If you do not register

your clicker, you will receive a zero for your clicker grade. Clicker points cannot be made up for any reason. If you forget/do not bring your clicker or are late and miss a clicker question, you miss those points for the day, regardless of the excuse.

If you click in for another student, or another student clicks in for you, or you are not physically present in the room when clicking in, this is considered academic dishonesty. You will receive a zero for your clicker quiz grade.

To accommodate illness, technical difficulties, etc., at least two weeks-worth of clicker points will be dropped at the end of the semester.

- [Getting Started with Clickers](#)
- [Using PointSolutions Mobile](#)

ACTIVITIES

There will be several activities this semester. These activities will account for 8% of your overall grade. There will be no make-ups for missed activities. To accommodate illness, technical difficulties, etc, at least one activity grade will be dropped at the end of the semester.

EXAMS

Four unit exams and a final exam are scheduled and will take place in the Math Department Testing Center, CHPN 327. The tentative testing dates are listed below and if changes are made you will be notified via email and on the homepage of the MLM course. Each unit exam is designed to be taken in 50 minutes or less. An additional 25 minutes has been added to the time allowed to accommodate typing in answers and/or the possibility of technical difficulties. Hence, you will be allowed a total of 75 minutes for each unit exam.

Before taking a unit exam, you are required to have completed the practice exam for that unit. The practice exam also counts as a quiz grade, so you'll want to do your best. If you score less than 70% on a unit exam, contact your instructor to set up a time when you can meet. This provides an opportunity for your instructor to help you formulate a plan for better preparing yourself for subsequent exams.

The final exam is comprehensive, required and is worth 20% of your overall course grade! If you do not take the final, you will fail the course regardless of your average.

The Math Department Testing Center hours of operation will be posted outside the Testing Center and on the website <http://mrtc.uark.edu/>. There are a limited number of computers in the testing center, and these are available on a first-come-first-served basis, so don't wait until the last minute.

The only things you need to bring to the Testing Lab are a pencil and your University ID. **You must show your university issued ID in order to take a test.** Scratch paper will be provided. You are not allowed any assistance of any kind on the tests and quizzes. Academic dishonesty is a violation of University's Academic Integrity Policy.

MAKE-UP EXAM POLICY

Make-up exams will be given only in **extreme** circumstances. If you need to miss an exam, you will need to contact your instructor **ahead of time**. **Documentation must be provided**. If you miss one exam, the “o” will be replaced with your grade from the final exam. If you do not miss any exams, your lowest unit exam score will be replaced with your grade from the final exam, provided this score is higher.

TENTATIVE TESTING DATES

	Days	Dates
Exam 1	Thursday or Friday	1/29/26 - 1/30/26
Exam 2	Thursday or Friday	2/19/26 – 2/20/26
Exam 3	Thursday or Friday	3/19/26 – 3/20/26
Exam 4	Wednesday or Thursday	4/22/26 – 4/23/26
Final Exam	Sunday or Monday	5/3/26 - 5/4/26

You are responsible for checking the Math Department Testing Center hours of operation and for making sure you arrive early enough to allow yourself the full amount of time to take your exam.

MATH DEPARTMENT TESTING CENTER POLICY

Make sure you read and understand the Math Department Testing Center Guidelines prior to taking the first exam in the Math Department Testing Center (CHPN 327). The Testing Center Guidelines may be viewed here: [Testing Center Guidelines](#)

COURSE GRADING

Homework	10%
Quizzes	10%
Clicker Grade	8%
Activities	8%
Exam 1	11%
Exam 2	11%
Exam 3	11%
Exam 4	11%
Final Exam	20%
TOTAL	100%

Assignment of letter grades is based on a percentage of points earned according to the following scale:

<i>A</i>	90% and above
<i>B</i>	80%-89.9%
<i>C</i>	70%-79.9%
<i>D</i>	60%-69.9%
<i>F</i>	0-59.9%

Your instructor is committed to keeping students informed of their standing in the class. Scores on all graded items will be posted in a timely manner. **Students are expected to bring any possible errors to the attention of the instructor within one week of posting.** This maintains an accurate grade record throughout the semester. **All scores posted on or before Reading Day will be deemed accurate at the time of the final exam.**

MEETING PREREQUISITES

The MLM gradebook can take up to 12 hours to update your scores. Therefore, you must have the required prerequisite assignments completed at least 12 hours before you plan to take a quiz or exam. Grades in MLM are approximations of your actual grade. Official grades will be kept in the MLM gradebook, NOT the Blackboard gradebook.

CLASSROOM ETIQUETTE

Students and instructors each have an important role in maintaining a classroom environment optimal for learning, are expected to treat each other with respect during class, using thoughtful dialogue, and keeping disruptive behaviors to a minimum. Both students and faculty perceive abusive language directed towards others as the most disruptive behavior. Other behaviors that can be disruptive are chatting and whispering during class, the use of smartphones or laptops for texting or in other ways unrelated to the course, preparing to leave before class is over, and consistently arriving late to class. Inappropriate behavior in the classroom may result in a request to immediately leave the classroom and/or a referral to the Office of Academic Integrity and Student Conduct.

ATTENDANCE POLICY

Students are expected to attend all classes; however, we understand that unexpected circumstances sometimes occur that may prevent you from making it to every class. Instead of “excusing” absences, we instead drop at least two weeks’ worth of clicker points to account for such circumstances. Should you be forced to miss an unusually high number of days, please contact your instructor as early as possible.

CONTACTING YOUR INSTRUCTOR

The best way to communicate with your instructor (beyond attending office hours) is by email. You can expect a reply within 24 hours, excluding weekends and holidays. Students are expected to monitor their UARK email address regularly and consult the course Blackboard site for important announcements.

You are encouraged to talk with your instructor if you have questions or need any help with this course. Contact information is located on the front page of this syllabus and in Blackboard. **When contacting your instructor, you must include your complete name as well the name of the class and the class meeting time.** Including these items in your message makes it much easier for your instructor to correspond with you. Please allow a minimum of 1 business day for a response from your instructor.

When communicating with your instructor, be sure to use appropriate language for an educational environment. Appropriate language includes using complete sentences and proper spelling and grammar. Uncommon abbreviations and/or obscene language should be avoided.

CONTINUITY PLAN

In the event of an extended campus closure, the continuity plan for this course includes the following:

- We will utilize our Blackboard course as the portal for the delivery of course materials and UARK email for communications. Please check both areas immediately for guidance and direction.
- We will utilize Blackboard Collaborate Ultra/Zoom to connect as a class during our regularly scheduled class days/times

- We will utilize Blackboard Collaborate Ultra/Zoom/Teams to connect during regularly scheduled office hours and by appointment.

ACADEMIC HONESTY STATEMENT

As a core part of its mission, the University of Arkansas provides students with the opportunity to further their educational goals through programs of study and research in an environment that promotes freedom of inquiry and academic responsibility. Accomplishing this mission is only possible when intellectual honesty and individual integrity prevail. Each University of Arkansas student is required to be familiar with and abide by the University's Academic Integrity Policy which may be found at honesty.uark.edu/policy. Students with questions about how these policies apply to a particular course or assignment should immediately contact their instructor.

ACCOMMODATIONS

Under University policy and federal and state law, students with documented disabilities are entitled to reasonable accommodations to ensure they have an equal opportunity to perform in class. If you have such a disability and need special academic accommodations, please report to Center for Educational Access (CEA). Reasonable accommodations may be arranged after CEA has verified your disability. You must communicate with your instructor to make sure any required accommodations have been put in place. Do not hesitate to contact your instructor if any assistance is needed in this process.

ADDITIONAL RESOURCES AND SERVICES

- **Free tutoring** is available in the Math Department Teaching Center (CHPN 326). The Math Department Teaching Center hours of operation can be found online at <http://mrtc.uark.edu/>.
- **Tutoring is also available through the Office of Student Success.** You can find more information at <https://success.uark.edu/tutors/chat.php#math>
- The [Pat Walker Health Center](#) has a variety of available health services, including on-campus clinics.
- [Counseling and Psychological Services \(C.A.P.S.\)](#) is available for students. Services are provided by licensed psychologists, counselors, and social workers.
- The [Center for Educational Access \(CEA\)](#) assists students with disabilities and will determine reasonable accommodations that will enable every student to access the full range of programs and services.

INCLEMENT WEATHER

Class will meet unless the University is closed. On-campus students are expected to be present. Off-campus students should make their own decisions in the best interest of personal safety. **If the university is officially closed, alternate methods of instruction will be communicated to you through a blackboard announcement and/or university email.** You will be notified of schedule changes for the Teaching and Testing Center in Champions Hall by email. If attendance is severely affected by weather, deadlines and exam dates may be adjusted. Please do not call the Department of Mathematical Sciences with weather-related inquiries. You may contact your instructor for more information.

UNAUTHORIZED USE OF TECHNOLOGY FOR A QUIZ OR EXAM

Students are not permitted to collaborate on any quiz or examination without specific permission from the instructor in advance. This includes collaboration through GroupMe, WhatsApp, or any other form of technology to exchange information associated with a quiz or examination.

The following is not all inclusive of what is considered academic misconduct for quizzes or examinations.

- Taking a screen shot of an online quiz or exam question, posting it to GroupMe or WhatsApp, and asking for assistance is considered academic misconduct.
- Answering an online quiz or exam question posted to GroupMe or WhatsApp is considered academic misconduct.
- Giving advice, assistance, or suggestions on how to complete a question associated with a quiz or examination is considered academic misconduct.
- The use of online websites (Quizlet, Chegg, Mathway, Wolfram Alpha, etc.) or search engines (Google) when exam instructions indicate these are not allowed is considered academic misconduct.
- Gathering to take an online quiz or exam with others and sharing answers in the process is considered academic misconduct.

Please note: If a student or group of students are found to be exchanging material associated with a quiz or examination through any form of technology (GroupMe, WhatsApp, etc.) or using any unauthorized resources (Googling answers, use of websites such as Quizlet, Course Hero, Chegg, etc.), I am required to report this matter per the University of Arkansas Academic Integrity Policy.

UNAUTHORIZED WEBSITES OR INTERNET RESOURCES

There are many websites claiming to offer study aids to students, but in using such websites, students could find themselves in violation of our University's Academic Integrity and Code of Student Life policies. These websites include (but are not limited to) Quizlet, Bartleby, Course Hero, Chegg, Wolfram Alpha, ChatGPT, and Clutch Prep. The U of A does not endorse the use of these products in an unethical manner. These websites may encourage students to upload course materials, such as test questions, individual assignments, and examples of graded material. Such materials are the intellectual property of instructors, the university, or publishers and may not be distributed without prior authorization. Furthermore, paying for academic work to be completed on your behalf and submitting it for academic credit is considered 'contract cheating' per the Academic Integrity Policy. Students found responsible for this type of violation face a grading penalty of 'XF' and a minimum one-semester academic suspension per the [University of Arkansas Sanction Rubric](#). Please let me know if you are uncertain about the use of a website.

UNAUTHORIZED USE OF CLASS RECORDINGS AND NOTES

Instructors may record class and make class available to students through Blackboard. These recordings may be used by students ONLY for the purposes of the class. Students may not download, store, copy, alter, post, share, or distribute in any manner all or any portion of the class recording, e.g. a 5-second clip of a class recording sent as a private message to one person is a violation of this provision. This provision may protect the following interests (as well as other interests not listed): faculty and university copyright; FERPA rights; and other privacy interests protected under state and/or federal law. Failure to comply with this provision will result in a referral to the [Office of Student Standards and Conduct](#) for potential charges under the [Code of Student Life](#). In situations where the recordings are used to gain an academic advantage, it may also be considered a violation of the [University of Arkansas' academic integrity policy](#).

Recording, or transmission of a recording, of all or any portion of a class is prohibited for students unless the recording is necessary for educational accommodation as expressly authorized and documented through the [Center for Educational Access](#) with proper advance

notice to the instructor. Unauthorized recordings may violate federal law, state law, and university policies. Student-made recordings are subject to the same restrictions as instructor-made recordings. Failure to comply with this provision will result in a referral to the [Office of Student Standards and Conduct](#) for potential charges under the [Code of Student Life](#). In situations where the recordings are used to gain an academic advantage, it may also be considered a violation of the [University of Arkansas' academic integrity policy](#).

Third parties may attempt to connect with you to buy your notes and other course information from this class. I will consider distributing course materials to a third party without my authorization a violation of my intellectual property rights and/or copyright law as well as a violation of the [University of Arkansas' academic integrity policy](#). Continued enrollment in this class signifies your intent to abide by the policy. Any violation will be reported to the [Office of Academic Initiatives and Integrity](#).

Please be aware that such class materials that may have already been given to such third parties may contain errors, which could affect your performance or grade. Recommendations for success in this course include coming to class on a routine basis, visiting me during my office hours, connecting with the Teaching Assistant (TA), and making use of [Student Success Center](#). If a third party should contact you regarding such an offer, I would appreciate your bringing this to my attention. We all play a part in creating a course climate of integrity.

EMERGENCY PROCEDURES

Many types of emergencies can occur on campus; instructions for specific emergencies such as severe weather, active shooter, or fire can be found at <http://emergency.uark.edu>.

Severe Weather (Tornado Warning):

- Follow the directions of the instructor or emergency personnel
- Seek shelter in the basement or interior room or hallway on the lowest floor, putting as many walls as possible between you and the outside
- If you are in a multi-story building, and you cannot get to the lowest floor, pick a hallway in the center of the building.
- Stay in the center of the room, away from exterior walls, windows, and doors.

Violence / Active Shooter (CADD):

- **CALL-** 9-1-1
- **AVOID-** If possible, self-evacuate to a safe area outside the building. Follow directions of police officers.
- **DENY-** Barricade the door with desk, chairs, bookcases or any items. Move to a place inside the room where you are not visible. Turn off the lights and remain quiet. Remain there until told by police its safe.
- **DEFEND-** Use chairs, desks, cell phones or whatever is immediately available to distract and/or defend yourself and others from attack.

NOTE

Your instructor reserves the right to make changes to the syllabus. You will be notified of any changes on the BB Announcement pages and/or via university email.

COURSE LEARNING OBJECTIVES

At the end of this course students will be able to demonstrate the following objectives.

Learning Objectives	Quiz	Exam
Evaluate limits using a graph	1.1 due 1/20/26	Exam 1 1/29/26 – 1/30/26
Evaluate limits using a table		
Evaluate one-side limits using limit laws and theorems		
Use given one-sided limits to find two-sided limits		
Apply properties of limits to find limits of given functions		
Use the properties of limits to evaluate infinite limits	1.2 Due 1/27/26	
Find points of discontinuity from a function definition		
Determine values of variables that make functions continuous		
Find left and right limits and points of discontinuity from a graph		
Find the average rate of change for a function		
Find an instantaneous rate of change for a function.	2.1 Due 2/10/26	Exam 2 2/19/26 – 2/20/26
Define the derivative as a rate of change		
Identify x-values where the derivative does not exist given the graph of the function.		
Apply the power rule for derivatives		
Find values where the slope of the tangent line has a given value.		
Write equations of tangent lines given a function and one point.	2.2 Due 2/17/26	
Use the product rule to find derivatives.		
Use the quotient rule to find derivatives.		
Use the chain rule to find derivatives.		
Use a combination of rules to find derivatives.		
Solve applications of the derivative.	3.1 due 3/3/26	Exam 3 3/19/26 – 3/20/26
Determine intervals on which a function is increasing or decreasing using a graph		
Determine critical numbers and where the functions are increasing or decreasing.		
Locate relative extrema from a given graph.		
Determine relative extrema algebraically using the first derivative test.		
Identify absolute extrema from a graph.	3.2 Due 3/10/26	
Find absolute extrema algebraically on a closed interval including applications.		
Find and evaluate higher order derivatives.		
Identify inflection points and concavity from a graph and algebraically.		
Find relative extrema using the second derivative test.		
Construct an objective function given a constraint.	3.3 Due 3/17/26	
Apply techniques for finding extrema to solve optimization applications.		
Find first and second order partial derivatives on multivariable functions.		
Identify critical points of multivariable functions.		
Identify relative extrema and saddle points of multivariable functions.		
Solve application problems involving partial derivatives.	4.1 4/7/26	Exam 4 4/22/26 – 4/23/26
Define the antiderivative.		
Find indefinite integrals using the power rule for integrals.		
Use U-substitution to find indefinite integrals		
Evaluate definite integrals using the Fundamental Theorem of Calculus		
Evaluate definite integrals involving U-substitution.	4.2 due 4/14/26	
Use areas to find definite integrals		
Use definite integrals to find areas.		
Find the area between two functions.		
Find the average value of a function over an interval including applications.		
Solve applications involving average value.	4.3 4/21/26	